

Problem Set 9

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1 Questions and Responses

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1. What is the dimension of the training data?
 - 404 observations and 14 predictors.
2. How many more X variables do you have than in the original housing data?
 - 61 additional predictors.

8 (LASSO)

1. What is the optimal value of λ ?
 - 0.002
2. What is the in-sample RMSE?
 - 0.138
3. What is the out-of-sample RMSE?
 - 0.184

9 (Ridge Regression)

1. What is the optimal value of λ ?
 - 0.037
2. What is the in-sample RMSE?
 - 0.140
3. What is the out-of-sample RMSE?
 - 0.180

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1. Would you be able to estimate a simple linear regression model on a data set that had more columns than rows?
 - No. If $p > n$, then $(X'X)$ is singular. Therefore, $\hat{\beta} = (X'X)^{-1}X'y$ is not defined, because $(X'X)^{-1}$ does not exist. Thus, you cannot estimate a unique OLS linear regression when $p > n$.
2. Using the RMSE values of each of the tuned models in the previous two questions, comment on where your model stands in terms of the bias-variance tradeoff.
 - Both models reduce overfitting, but ridge achieves the slightly better bias-variance tradeoff here because it has the lower test RMSE. LASSO fits the training data slightly better, while ridge generalizes slightly better. This suggests LASSO has slightly lower bias and higher variance, whereas ridge accepts slightly more bias in exchange for lower variance.